# Inheritance

60 questions: 30 multiple choice, 30 structured.

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## 1 · Part 1 Core Chromosomes, genes and proteins [1]

What material forms chromosomes and carries genetic information?

A starch

B DNA

C fat

D cellulose

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## 2 · Part 1 Core Chromosomes, genes and proteins [1]

Which statement defines a gene?

A a whole cell containing two nuclei

B any protein found in a chromosome

C a membrane surrounding a nucleus

D a length of DNA that codes for a protein

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## 3 · Part 1 Core Chromosomes, genes and proteins [1]

Two versions of one gene occur at the same position on corresponding chromosomes. What are these versions called?

A alleles

B ribosomes

C gametes

D enzymes

## 4 · Part 1 Core Chromosomes, genes and proteins [1]

Which normal combination of sex chromosomes occurs in a human male body cell?

A XX

B YY

C XY

D X alone

## 5 · Part 1 Core Chromosomes, genes and proteins [1]

An egg containing an X chromosome is fertilised by a sperm containing an X chromosome. What is the expected sex-chromosome combination in the zygote?

A XY

B YY

C XXX

D XX

## 6 · Part 1 Supplement Chromosomes, genes and proteins [1]

Which feature of a gene most directly determines the amino-acid sequence of its protein product?

A the position of the cell in a tissue

B the order of its bases

C the amount of starch in the cell

D the colour of the nucleus

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## 7 · Part 1 Supplement Chromosomes, genes and proteins [1]

Two proteins contain different amino-acid sequences. Which consequence is most directly expected?

A They must contain identical active sites.

B They must be DNA molecules.

C They may fold into different shapes.

D They must have the same function.

## 8 · Part 1 Supplement Chromosomes, genes and proteins [1]

A cell makes a different membrane carrier after a change in a gene. How can this affect cell function?

A It can alter the movement of a substance across the membrane.

B It changes the membrane into a chromosome.

C It prevents the cell containing any DNA.

D It turns every cell protein into the same carrier.

## 9 · Part 1 Supplement Chromosomes, genes and proteins [1]

Which molecule carries a copy of a gene's information from the nucleus to the cytoplasm?

A cellulose

B mRNA

C glycogen

D a whole chromosome leaving the nucleus

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## 10 · Part 1 Supplement Chromosomes, genes and proteins [1]

Where are amino acids assembled into a protein during protein synthesis?

A cell wall

B vacuole membrane

C ribosome

D chromosome surface in the nucleus

## 11 · Part 1 Supplement Chromosomes, genes and proteins [1]

A skin cell and a muscle cell from the same person make different proteins. Which explanation is best?

A All their genes must be different.

B The muscle cell has no chromosomes.

C The skin cell has no DNA.

D Different genes are expressed in the two cell types.

## 12 · Part 1 Supplement Chromosomes, genes and proteins [1]

A nucleus contains one complete set of chromosomes. How is it described?

A haploid

B diploid

C a protein

D a pair of gametes

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## 13 · Part 1 Supplement Chromosomes, genes and proteins [1]

A diploid nucleus has which arrangement?

A one chromosome in total

B no genetic material

C two sets of chromosomes

D one set of amino acids

## 14 · Part 1 Supplement Chromosomes, genes and proteins [1]

How many chromosomes are normally present in a human diploid body-cell nucleus?

A 46

B 23

C 44

D 92

## 15 · Part 1 Supplement Chromosomes, genes and proteins [1]

During protein synthesis, what determines the order in which a ribosome assembles amino acids?

A the order of cells in the tissue

B the base sequence of the mRNA passing through it

C the concentration of oxygen alone

D the position of the ribosome in the cell membrane

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## 16 · Part 1 Core Chromosomes, genes and proteins [4]

An inherited condition is associated with a gene on a chromosome.

(a) State what chromosomes are made of and define a gene. [2]

(b) Define an allele and explain why two alleles of this gene can differ. [2]

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## 17 · Part 1 Core Chromosomes, genes and proteins [4]

The parents of a child have normal human sex-chromosome combinations.

(a) State the sex chromosomes in each parent's body cells. [2]

(b) Explain why either an XX or an XY zygote can be produced. [2]

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## 18 · Part 1 Supplement Chromosomes, genes and proteins [4]

Two alleles produce proteins with different shapes.

(a) Describe the link between DNA and the amino-acid sequence. [2]

(b) Explain how this difference can affect an enzyme's activity. [2]

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**19 · Part 1** Supplement Chromosomes, genes and proteins [4]

A cell's gene controls production of an enzyme involved in one metabolic reaction.

(a) Explain why this gene can affect the cell's metabolism. [2]

(b) State two other kinds of protein through which DNA can control cell function. [2]

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## 20 · Part 1 Supplement Chromosomes, genes and proteins [4]

A cell begins making a protein coded by one of its nuclear genes.

(a) Describe the events that carry the information to the cytoplasm. [2]

(b) Describe how the information is used there. [2]

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## 21 · Part 1 Supplement Chromosomes, genes and proteins [4]

A pancreatic cell makes a protein that a nerve cell from the same person does not make.

(a) Explain why this does not mean that the gene is absent from the nerve cell. [2]

(b) Explain how gene expression contributes to cell specialisation. [2]

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## 22 · Part 1 Supplement Chromosomes, genes and proteins [4]

A plant has 16 chromosomes in a normal diploid body-cell nucleus.

(a) Define diploid and give the chromosome number in a haploid nucleus of this plant. [2]

(b) Explain why the haploid nucleus has this number. [2]

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## 23 · Part 1 Supplement Chromosomes, genes and proteins [4]

A laboratory compares normal human body-cell and gamete nuclei.

(a) State the chromosome number in each type of nucleus. [2]

(b) Describe their chromosome sets using the terms haploid and diploid. [2]

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## 24 · Part 1 Supplement Chromosomes, genes and proteins [4]

A student claims that a gene leaves the nucleus and becomes a protein.

(a) Correct the claim about what leaves the nucleus. [2]

(b) Correct the claim about how the protein is formed. [2]

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**25 · Part 1** Supplement Chromosomes, genes and proteins [4]

A receptor protein is abundant in one cell type but absent in another cell type of the same organism.

(a) Explain how DNA can control the response to a chemical signal. [2]

(b) Explain why the two cell types can differ despite containing the same genes. [2]

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## 26 · Part 1 Core Chromosomes, genes and proteins [4]

In a family, the first two children have normal XX sex chromosomes.

(a) State the two possible sex chromosomes in the father's sperm. [2]

(b) Explain why this history does not make an XY next child certain. [2]

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## 27 · Part 1 Supplement Chromosomes, genes and proteins [4]

A protein used as a membrane carrier has a different amino-acid order in two individuals.

(a) State the genetic feature that specifies this order and suggest why it differs between the two individuals. [2]

(b) Explain why the carrier might not work equally well in both individuals. [2]

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## 28 · Part 1 Supplement Chromosomes, genes and proteins [4]

A protein contains the amino acids alanine, glycine and serine in a specific order.

(a) Explain why a supply of these amino acids alone does not specify the correct protein. [2]

(b) Describe the role of mRNA and the ribosome in obtaining the correct sequence. [2]

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**29 · Part 1** Supplement Chromosomes, genes and proteins [4]

Two species have different normal diploid chromosome numbers: species P has 10 and species Q has 28.

(a) Calculate their haploid chromosome numbers. [2]

(b) Explain why a larger chromosome number does not change the meaning of diploid. [2]

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## 30 · Part 1 Supplement Chromosomes, genes and proteins [4]

A change in one gene reduces the amount of a particular enzyme made in a cell.

(a) Describe the relationship between chromosome, DNA and gene. [2]

(b) Explain how a change in this gene can alter the cell's function. [2]

## 1 · Part 2 Supplement Mitosis [1]

Which outcome describes mitosis in a diploid body cell?

A daughter nuclei with half the chromosome number

B daughter nuclei with the same genetic information as the parent nucleus

C fusion of two gamete nuclei

D offspring nuclei receiving no parental DNA

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## 2 · Part 2 Supplement Mitosis [1]

What must happen to chromosomes before mitosis?

A They are all destroyed.

B They fuse with chromosomes from a sperm.

C Their number is permanently halved.

D They are replicated exactly.

## 3 · Part 2 Supplement Mitosis [1]

Which description applies to a stem cell?

A an unspecialised cell whose daughter cells can become specialised

B a cell that cannot divide

C a cell that contains no chromosomes

D a cell that can form only pollen grains

## 4 · Part 2 Supplement Meiosis [1]

Which process is involved in producing gametes?

A fertilisation

B pollination

C meiosis

D digestion

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## 5 · Part 2 Supplement Meiosis [1]

A diploid cell with 32 chromosomes undergoes meiosis. What is expected in the resulting gamete nuclei?

A. 32 chromosomes and identical genetic information

B. 64 chromosomes and no genetic variation

C. 8 chromosomes and identical genetic information

D. 16 chromosomes and genetic differences between nuclei

## 6 · Part 2 Core Monohybrid inheritance [1]

A plant has genotype Rr and red flowers. Which term describes its red flower colour?

A. genotype

B. phenotype

C. allele

D. gamete

## 7 · Part 2 Core Monohybrid inheritance [1]

Which description applies to a heterozygous individual?

A. It has two identical dominant alleles of a gene.

B. It has two identical recessive alleles of a gene.

C. It has two different alleles of a gene.

D. It has only one allele of a gene.

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## 8 · Part 2 Core Monohybrid inheritance [1]

Allele B is dominant to b. Which genotype expresses the recessive feature?

A bb

B BB

C Bb

D both Bb and bb

## 9 · Part 2 Core Monohybrid inheritance [1]

In a plant, P gives purple flowers and is dominant to p. What phenotypic ratio is expected from Pp crossed with pp?

A 3 purple : 1 white

B 1 purple : 1 white

C 1 purple : 3 white

D all purple

## 10 · Part 2 Core Monohybrid inheritance [1]

A cross between two heterozygous animals gives a 3 : 1 dominant-to-recessive phenotypic ratio. Which offspring genotype expresses the recessive phenotype?

A the heterozygous genotype

B the homozygous dominant genotype

C the homozygous recessive genotype

D every genotype produced

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## 11 · Part 2 Supplement Monohybrid inheritance [1]

To find whether a plant with a dominant phenotype is homozygous, which partner is used in a test cross?

A a homozygous dominant plant only

B any plant with the same dominant phenotype

C a plant from another species

D a homozygous recessive plant

## 12 · Part 2 Supplement Monohybrid inheritance [1]

A heterozygote shows contributions from both alleles in its phenotype. What is this called?

A codominance

B complete dominance of one allele

C absence of both alleles

D asexual reproduction

## 13 · Part 2 Supplement Monohybrid inheritance [1]

Which genotype gives blood group O?

A $I^AI^o$

B $I^BI^o$

C $I^oI^o$

D $I^AI^B$

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## 14 · Part 2 Supplement Monohybrid inheritance [1]

Why can red-green colour blindness be more common in males than females?

A The relevant recessive allele is on the X chromosome and males normally have only one X.

B Males normally have two X chromosomes.

C The allele is present only in mitochondria.

D Every allele on an X chromosome is dominant.

## 15 · Part 2 Core Monohybrid inheritance [1]

Under complete dominance, which mating is expected to be pure-breeding for the recessive feature?

A Aa crossed with Aa

B aa crossed with aa

C AA crossed with Aa

D Aa crossed with aa

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## 16 · Part 2 Supplement Mitosis [4]

A tissue replaces damaged cells by mitosis. Its normal cells have 24 chromosomes.

(a) State the chromosome number in each daughter nucleus and explain why it is maintained. [2]

(b) State two other roles of mitosis. [2]

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## 17 · Part 2

Supplement

Mitosis

[4]

Before one body cell divides, its chromosomes are copied.

(a) Explain why exact copying is necessary. [2]

(b) Describe what happens to the chromosome copies during mitosis. [2]

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## 18 · Part 2 Supplement Mitosis [4]

Stem cells are used to replace some damaged specialised cells.

(a) Describe two features of stem cells that make this possible. [2]

(b) Explain what must happen to some daughter cells before they perform the damaged tissue's function. [2]

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## 19 · Part 2 Supplement Meiosis [4]

A species has 12 chromosomes in its diploid nuclei.

(a) Describe the chromosome number and genetic similarity of gametes produced by meiosis. [2]

(b) Explain why meiosis is called a reduction division. [2]

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## 20 · Part 2 Core Monohybrid inheritance [4]

A breeder records that a plant has genotype Hh and hairy leaves.

(a) Explain the terms genotype and phenotype using this plant. [2]

(b) Describe what is meant by inheritance and state what is passed between generations. [2]

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## 21 · Part 2 Core Monohybrid inheritance [4]

In a species, black coat allele B is dominant to white coat allele b.

(a) Identify which of BB, Bb and bb are homozygous and which is heterozygous. [2]

(b) Explain the coat colours of Bb and bb animals. [2]

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## 22 · Part 2 Core Monohybrid inheritance [4]

In a seed species, smooth allele S is dominant to wrinkled allele s. Two Ss plants are crossed.

(a) State the gametes from each parent and the possible offspring genotypes. [2]

(b) State the expected phenotypic ratio and explain the recessive category. [2]

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## 23 · Part 2 Core Monohybrid inheritance [4]

In a fly species, long-wing allele L is dominant to short-wing allele l. A Ll fly is crossed with a ll fly.

(a) State the possible gametes from each parent. [2]

(b) Determine the possible offspring genotypes and phenotypic ratio. [2]

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## 24 · Part 2 Supplement Monohybrid inheritance [4]

A plant has red flowers. Allele R is dominant to r, which gives white flowers. Its genotype is unknown.

(a) Describe a suitable test cross. [2]

(b) Explain what white-flowered offspring would show and why a small all-red sample is inconclusive. [2]

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## 25 · Part 2 Supplement Monohybrid inheritance [4]

In a cattle population, allele $C^R$ gives red hair and $C^W$ gives white hair. Heterozygotes have both red and white hairs. Two heterozygotes are crossed.

(a) State the gametes and the possible offspring genotypes. [2]

(b) Determine the phenotypic ratio and explain the heterozygote's appearance. [2]

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## 26 · Part 2 Supplement Monohybrid inheritance [4]

One parent has blood group A with genotype $I^AI^o$. The other has blood group B with genotype $I^BI^o$.

(a) State the possible gametes from each parent. [2]

(b) Determine the possible children's blood groups and their expected ratio. [2]

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## 27 · Part 2 Supplement Monohybrid inheritance [4]

For red-green colour blindness, $X^N$ carries the normal allele and $X^n$ the recessive allele. A carrier woman $X^NX^n$ and a normal-vision man $X^NY$ have children.

(a) State the sex-chromosome combinations possible in their daughters. [2]

(b) Determine the proportion of their sons expected to be colour blind and explain. [2]

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## 28 · Part 2 Core Monohybrid inheritance [4]

Purple allele P is dominant to white allele p. A gardener repeatedly crosses plants from a line believed to breed true for purple flowers.

(a) State one genotype consistent with a pure-breeding purple line and explain. [2]

(b) Explain why a Pp plant does not breed true when crossed with another Pp plant. [2]

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## 29 · Part 2 Supplement Monohybrid inheritance [4]

A person with blood group AB has children with a person with blood group O.

(a) State the parental genotypes and their gametes. [2]

(b) Determine the expected blood groups and explain why AB is not expected. [2]

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## 30 · Part 2 Core Monohybrid inheritance [4]

In a monohybrid cross, two heterozygous parents produce 200 offspring. Dominance is complete and all genotypes survive equally.

(a) Calculate the expected numbers with dominant and recessive phenotypes. [2]

(b) Explain why the observed numbers may differ from these expectations. [2]

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